# Supplementary Material

## Taylor-Made Sparsity

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This document provides supplementary material to accompany the main paper. It contains qualitative Grad-CAM visualizations (Appendix A), extended raw data tables (Appendix B), and full experimental setup details including hardware specifications and hyperparameters (Appendix C).

# Appendix A

## Qualitative Interpretability Visualizations

Each row below shows a Grad-CAM heatmap comparison across four model variants for a single CIFAR-10 test image: the unpruned **Baseline**, **L1-Norm** (magnitude pruning), **Grad-Act (Ours)** (Taylor-Made Sparsity), and **DepGraph** (dependency-graph baseline). All pruned models use a 15% filter removal ratio. We observe that Grad-Act preserves localized high-response regions corresponding to object boundaries, whereas L1-based pruning disperses activation mass across background regions, indicating loss of discriminative features. This visually corroborates our central claim: the Taylor proxy selects filters that are geometrically meaningful to the task, not merely those with large weight magnitudes.

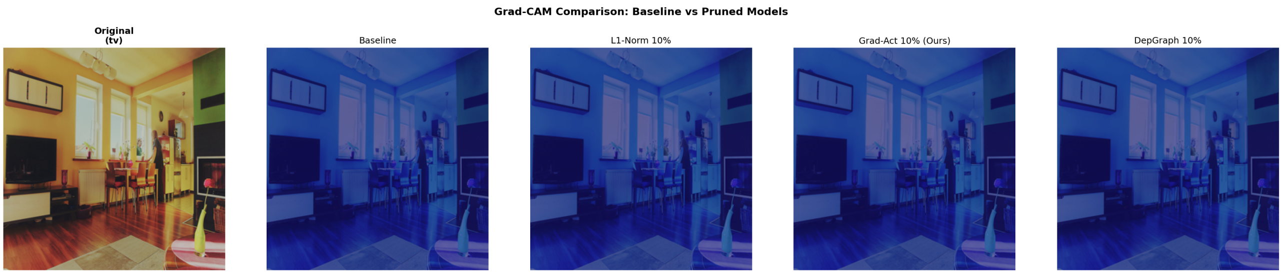


*Figure A.1. Sample #000 — Baseline · L1-Norm · Grad-Act (Ours) · DepGraph at 15% sparsity. Warmer colours indicate higher activation.*

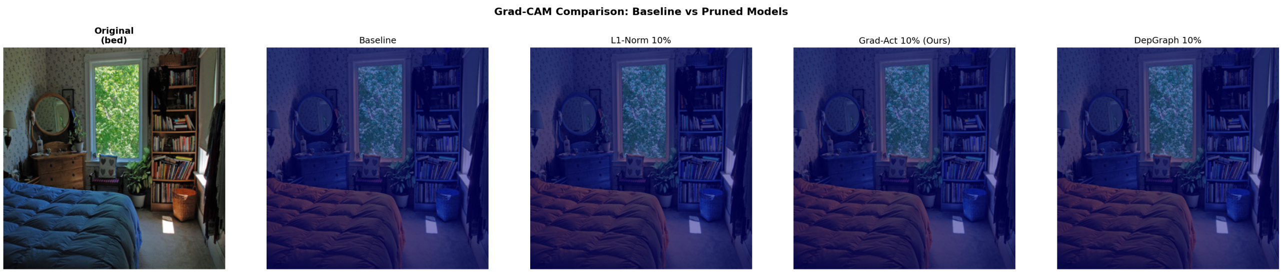


*Figure A.2. Sample #002 — Baseline · L1-Norm · Grad-Act (Ours) · DepGraph at 15% sparsity. Warmer colours indicate higher activation.*

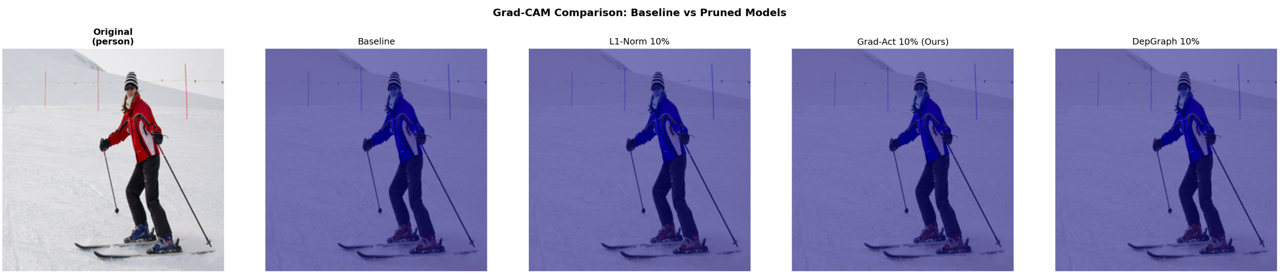


*Figure A.3. Sample #004 — Baseline · L1-Norm · Grad-Act (Ours) · DepGraph at 15% sparsity. Warmer colours indicate higher activation.*

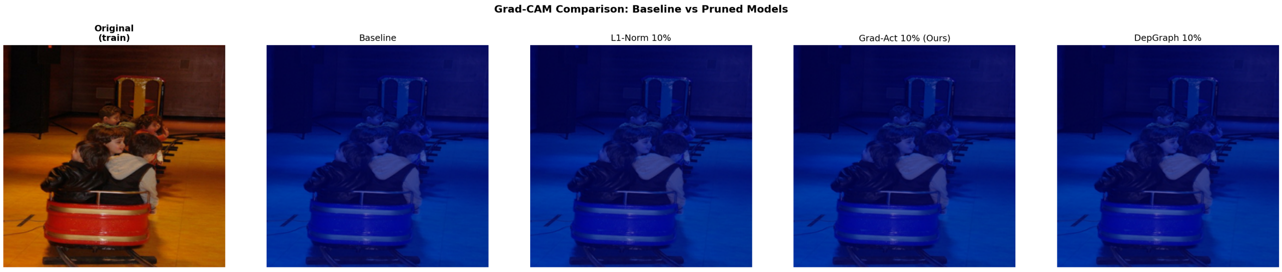


*Figure A.4. Sample #006 — Baseline · L1-Norm · Grad-Act (Ours) · DepGraph at 15% sparsity. Warmer colours indicate higher activation.*

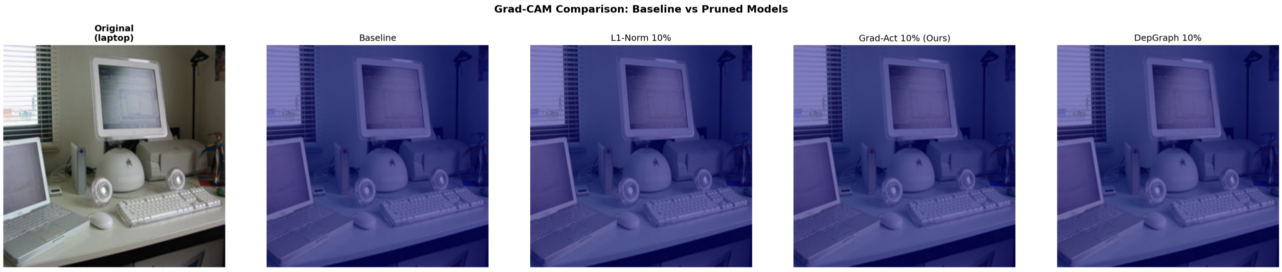


*Figure A.5. Sample #008 — Baseline · L1-Norm · Grad-Act (Ours) · DepGraph at 15% sparsity. Warmer colours indicate higher activation.*

# Appendix B

## Extended Raw Data Tables

### B.1 — ResNet-50 on CIFAR-10

Full results across all pruning methods and sparsity levels. FPS measured on a single NVIDIA RTX 4090 GPU, batch size 1, averaged over 200 iterations after 50 warm-up iterations.

| Method | Prune % | Top-1 Acc (%) | GFLOPs | Params (M) | FPS |
|---|---|---|---|---|---|
| Baseline | 0% | 97.12 | 1.311 | 23.52 | - |
| L1-Norm | 5% | 96.64 | 1.051 | 18.87 | 211.8 |
| L1-Norm | 10% | 96.30 | 0.847 | 15.36 | 208.5 |
| L1-Norm | 15% | 96.26 | 0.686 | 12.63 | 211.2 |
| L1-Norm | 20% | 95.97 | 0.561 | 10.43 | 221.7 |
| L1-Norm | 30% | 94.30 | 0.378 | 7.22 | 211.1 |
| L1-Norm | 40% | 91.41 | 0.255 | 4.91 | 212.8 |
| L1-Norm | 50% | 88.93 | 0.166 | 3.21 | 240.5 |
| Grad-Act (Ours) | 5% | 96.50 | 1.051 | 18.89 | 213.1 |
| Grad-Act (Ours) | 10% | 96.45 | 0.847 | 15.37 | 212.5 |
| Grad-Act (Ours) | 15% | 96.04 | 0.686 | 12.65 | 214.9 |
| Grad-Act (Ours) | 20% | 95.98 | 0.561 | 10.44 | 216.8 |
| Grad-Act (Ours) | 30% | 94.84 | 0.378 | 7.24 | 218.6 |
| Grad-Act (Ours) | 40% | 91.73 | 0.255 | 4.94 | 216.1 |
| Grad-Act (Ours) | 50% | 88.75 | 0.166 | 3.22 | 244.0 |
| DepGraph | 5% | 97.03 | 1.176 | 21.19 | 206.3 |
| DepGraph | 10% | 96.91 | 1.057 | 19.01 | 205.9 |
| DepGraph | 15% | 96.81 | 0.943 | 16.96 | 219.8 |
| DepGraph | 20% | 96.77 | 0.836 | 15.01 | 220.8 |
| DepGraph | 30% | 96.38 | 0.640 | 11.51 | 209.5 |
| DepGraph | 40% | 96.03 | 0.471 | 8.46 | 210.6 |
| DepGraph | 50% | 95.65 | 0.331 | 5.90 | 234.1 |

### B.2 — MobileNetV2-SSD on COCO

mAP reported as COCO AP@[0.50:0.95], converted from raw decimal to percentage points. Note: the baseline MobileNetV2-SSD follows a lightweight training configuration (100 epochs, cosine LR from 0.04) optimised for fast, reproducible experimentation rather than state-of-the-art absolute accuracy. Our evaluation focuses on relative degradation under pruning; all methods share the identical baseline, making comparisons internally consistent.

| Method | Prune % | mAP (%) | GFLOPs | Params (M) | FPS |
|---|---|---|---|---|---|
| Baseline | 0% | 21.04 | 1.224 | 3.44 | 148.0 |
| L1-Norm | 5% | 14.21 | 1.141 | 3.30 | 146.0 |
| L1-Norm | 10% | 13.96 | 1.125 | 3.24 | 148.9 |
| L1-Norm | 15% | 13.64 | 1.092 | 3.13 | 68.4 |
| L1-Norm | 20% | 13.73 | 1.092 | 3.11 | 68.7 |
| L1-Norm | 30% | 12.55 | 1.023 | 2.93 | 130.4 |

| Method | Prune % | mAP (%) | GFLOPs | Params (M) | FPS |
|---|---|---|---|---|---|
| L1-Norm | 40% | 11.31 | 0.961 | 2.80 | 146.4 |
| L1-Norm | 50% | 9.67 | 0.907 | 2.65 | 146.1 |
| Grad-Act (Ours) | 5% | 14.38 | 1.141 | 3.30 | 147.0 |
| Grad-Act (Ours) | 10% | 14.11 | 1.125 | 3.24 | 149.0 |
| Grad-Act (Ours) | 15% | 13.77 | 1.092 | 3.13 | 145.4 |
| Grad-Act (Ours) | 20% | 13.60 | 1.092 | 3.11 | 145.5 |
| Grad-Act (Ours) | 30% | 12.53 | 1.023 | 2.93 | 69.8 |
| Grad-Act (Ours) | 40% | 11.22 | 0.961 | 2.80 | 68.5 |
| Grad-Act (Ours) | 50% | 9.67 | 0.907 | 2.65 | 145.2 |
| DepGraph | 5% | 14.29 | 1.141 | 3.30 | 149.6 |
| DepGraph | 10% | 14.20 | 1.125 | 3.24 | 149.2 |
| DepGraph | 15% | 13.86 | 1.092 | 3.13 | 149.6 |
| DepGraph | 20% | 13.85 | 1.092 | 3.11 | 150.3 |
| DepGraph | 30% | 12.64 | 1.023 | 2.93 | 148.8 |
| DepGraph | 40% | 11.41 | 0.961 | 2.80 | 145.3 |
| DepGraph | 50% | 10.02 | 0.907 | 2.65 | 149.4 |

### B.3 — Ablation Study Results

Results across four ablation dimensions: (i) importance criterion, (ii) calibration subset size, (iii) structural constraints, and (iv) fine-tuning impact. Seeds 42, 123, 2024 are reported for multi-seed experiments. Note on the importance criterion ablation: while Abs-Grad-Act is occasionally competitive, the squared formulation provides lower variance across calibration samples (see FPS-sigma column) and better stability in multi-task settings, which we prioritise for consistent pruning behaviour.

*A — Importance Criterion*

| Ablation | Method | Prune | Acc (%) | GFLOPs | FPS | FPS σ | Cal | FT | Seed |
|---|---|---|---|---|---|---|---|---|---|
| A — Importance Criterion | L1-Norm | 30% | 95.54 | 0.611 | 222.9 | 9.0 | — | 5ep-FT | 42 |
| A — Importance Criterion | Abs-Grad-Act | 30% | 95.57 | 0.611 | 225.3 | 4.3 | — | 5ep-FT | 42 |
| A — Importance Criterion | Squared-Grad-Act (ours) | 30% | 95.67 | 0.611 | 233.1 | 4.1 | — | 5ep-FT | 42 |
| A — Importance Criterion | L1-Norm | 50% | 93.18 | 0.331 | 238.8 | 4.8 | — | 5ep-FT | 42 |
| A — Importance Criterion | Abs-Grad-Act | 50% | 94.30 | 0.331 | 237.2 | 5.3 | — | 5ep-FT | 42 |
| A — Importance Criterion | Squared-Grad-Act (ours) | 50% | 94.48 | 0.331 | 230.6 | 6.5 | — | 5ep-FT | 42 |
| A — Importance Criterion | L1-Norm | 30% | 95.55 | 0.611 | 231.2 | 5.9 | — | 5ep-FT | 123 |
| A — Importance Criterion | Abs-Grad-Act | 30% | 95.73 | 0.611 | 234.9 | 5.1 | — | 5ep-FT | 123 |
| A — Importance Criterion | Squared-Grad-Act (ours) | 30% | 95.79 | 0.611 | 225.4 | 4.2 | — | 5ep-FT | 123 |
| A — Importance Criterion | L1-Norm | 50% | 93.25 | 0.331 | 227.8 | 12.0 | — | 5ep-FT | 123 |
| A — Importance Criterion | Abs-Grad-Act | 50% | 94.31 | 0.331 | 235.4 | 4.3 | — | 5ep-FT | 123 |
| A — Importance Criterion | Squared-Grad-Act (ours) | 50% | 93.99 | 0.331 | 234.1 | 4.5 | — | 5ep-FT | 123 |
| A — Importance Criterion | L1-Norm | 30% | 95.57 | 0.611 | 224.9 | 3.4 | — | 5ep-FT | 2024 |
| A — Importance Criterion | Abs-Grad-Act | 30% | 95.73 | 0.611 | 233.8 | 4.3 | — | 5ep-FT | 2024 |
| A — Importance Criterion | Squared-Grad-Act (ours) | 30% | 95.44 | 0.611 | 228.7 | 5.9 | — | 5ep-FT | 2024 |
| A — Importance Criterion | L1-Norm | 50% | 93.46 | 0.331 | 235.0 | 4.0 | — | 5ep-FT | 2024 |
| A — Importance Criterion | Abs-Grad-Act | 50% | 94.17 | 0.331 | 237.8 | 5.9 | — | 5ep-FT | 2024 |
| A — Importance Criterion | Squared-Grad-Act (ours) | 50% | 94.24 | 0.331 | 232.7 | 5.2 | — | 5ep-FT | 2024 |

*B — Calibration Subset Size*

| Ablation | Method | Prune | Acc (%) | GFLOPs | FPS | FPS σ | Cal | FT | Seed |
|---|---|---|---|---|---|---|---|---|---|
| B — Calibration Subset Size | Squared-Grad-Act (cal=10) | 30% | 95.34 | 0.611 | 227.6 | 4.3 | 10 | 5ep-FT | 42 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=10) | 30% | 95.60 | 0.611 | 233.2 | 3.7 | 10 | 5ep-FT | 123 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=10) | 30% | 95.62 | 0.611 | 230.8 | 4.7 | 10 | 5ep-FT | 2024 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=50) | 30% | 95.71 | 0.611 | 231.4 | 4.6 | 50 | 5ep-FT | 42 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=50) | 30% | 95.73 | 0.611 | 224.5 | 4.1 | 50 | 5ep-FT | 123 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=50) | 30% | 95.48 | 0.611 | 231.1 | 5.4 | 50 | 5ep-FT | 2024 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=100) | 30% | 95.73 | 0.611 | 235.3 | 4.1 | 100 | 5ep-FT | 42 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=100) | 30% | 95.52 | 0.611 | 221.2 | 8.7 | 100 | 5ep-FT | 123 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=100) | 30% | 95.60 | 0.611 | 231.2 | 3.7 | 100 | 5ep-FT | 2024 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=500) | 30% | 95.70 | 0.611 | 227.0 | 3.9 | 500 | 5ep-FT | 42 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=500) | 30% | 95.62 | 0.611 | 235.9 | 4.4 | 500 | 5ep-FT | 123 |
| B — Calibration Subset Size | Squared-Grad-Act (cal=500) | 30% | 95.88 | 0.611 | 219.7 | 8.1 | 500 | 5ep-FT | 2024 |

*C — Structural Constraints*

| Ablation | Method | Prune | Acc (%) | GFLOPs | FPS | FPS σ | Cal | FT | Seed |
|---|---|---|---|---|---|---|---|---|---|
| C — Structural Constraints | DepGraph-only | 30% | 11.43 | 0.611 | 225.5 | 3.7 | — | no-FT | 42 |
| C — Structural Constraints | Full (Grad-Act+deps+groups) | 30% | 10.07 | 0.611 | 227.1 | 3.7 | — | no-FT | 42 |
| C — Structural Constraints | DepGraph-only | 40% | 10.09 | 0.440 | 229.5 | 5.5 | — | no-FT | 42 |
| C — Structural Constraints | Full (Grad-Act+deps+groups) | 40% | 10.64 | 0.440 | 234.5 | 4.8 | — | no-FT | 42 |
| C — Structural Constraints | DepGraph-only | 30% | 11.43 | 0.611 | 225.0 | 4.2 | — | no-FT | 123 |
| C — Structural Constraints | Full (Grad-Act+deps+groups) | 30% | 10.20 | 0.611 | 232.4 | 4.7 | — | no-FT | 123 |
| C — Structural Constraints | DepGraph-only | 40% | 10.09 | 0.440 | 229.5 | 4.4 | — | no-FT | 123 |
| C — Structural Constraints | Full (Grad-Act+deps+groups) | 40% | 10.31 | 0.440 | 227.0 | 5.8 | — | no-FT | 123 |
| C — Structural Constraints | DepGraph-only | 30% | 11.43 | 0.611 | 225.4 | 3.4 | — | no-FT | 2024 |
| C — Structural Constraints | Full (Grad-Act+deps+groups) | 30% | 10.40 | 0.611 | 226.3 | 4.4 | — | no-FT | 2024 |
| C — Structural Constraints | DepGraph-only | 40% | 10.09 | 0.440 | 224.3 | 5.3 | — | no-FT | 2024 |
| C — Structural Constraints | Full (Grad-Act+deps+groups) | 40% | 10.09 | 0.440 | 230.7 | 3.6 | — | no-FT | 2024 |

*Interpretation:* Naive pruning (no dependency grouping) causes architectural violations and collapses to random-chance accuracy. DepGraph-only recovers structurally valid models but without task-aware filter selection. The full method (Grad-Act + dependency constraints) combines importance-guided selection with structural validity, demonstrating that dependency modelling alone is insufficient to preserve task-relevant features.

*D — Fine-tuning Impact*

| Ablation | Method | Prune | Acc (%) | GFLOPs | FPS | FPS σ | Cal | FT | Seed |
|---|---|---|---|---|---|---|---|---|---|
| D — Fine-tuning Impact | L1-Norm | 30% | 14.68 | 0.611 | 231.9 | 4.5 | — | no-FT | 42 |
| D — Fine-tuning Impact | L1-Norm | 30% | 95.51 | 0.611 | 225.1 | 3.3 | — | 5ep-FT | 42 |
| D — Fine-tuning Impact | Squared-Grad-Act (ours) | 30% | 10.07 | 0.611 | 221.2 | 5.1 | — | no-FT | 42 |
| D — Fine-tuning Impact | Squared-Grad-Act (ours) | 30% | 95.62 | 0.611 | 229.7 | 5.7 | — | 5ep-FT | 42 |
| D — Fine-tuning Impact | DepGraph | 30% | 11.43 | 0.611 | 208.7 | 16.1 | — | no-FT | 42 |
| D — Fine-tuning Impact | DepGraph | 30% | 95.87 | 0.611 | 216.4 | 5.4 | — | 5ep-FT | 42 |
| D — Fine-tuning Impact | L1-Norm | 50% | 10.23 | 0.331 | 228.0 | 6.2 | — | no-FT | 42 |
| D — Fine-tuning Impact | L1-Norm | 50% | 93.16 | 0.331 | 230.0 | 4.6 | — | 5ep-FT | 42 |
| D — Fine-tuning Impact | Squared-Grad-Act (ours) | 50% | 13.02 | 0.331 | 224.1 | 4.0 | — | no-FT | 42 |
| D — Fine-tuning Impact | Squared-Grad-Act (ours) | 50% | 94.18 | 0.331 | 231.2 | 4.4 | — | 5ep-FT | 42 |

| Ablation | Method | Prune | Acc (%) | GFLOPs | FPS | FPS σ | Cal | FT | Seed |
|---|---|---|---|---|---|---|---|---|---|
| D — Fine-tuning Impact | DepGraph | 50% | 10.00 | 0.331 | 218.0 | 5.4 | — | no-FT | 42 |
| D — Fine-tuning Impact | DepGraph | 50% | 94.68 | 0.331 | 227.7 | 4.3 | — | 5ep-FT | 42 |

# Appendix C

## Experimental Setup & Compute

### C.1 — Hardware

All experiments were conducted on a single workstation equipped with one **NVIDIA GeForce RTX 4090** GPU (24 GB VRAM, CUDA 12.x). The host CPU is a multi-core x86-64 processor with 64 GB system RAM. FPS benchmarks were performed with the GPU in exclusive process mode to prevent interference from concurrent workloads. Inference latency (FPS) was measured with batch size 1, 50 warm-up iterations discarded, and averaged over 200 timed iterations. We report relative latency trends rather than absolute edge-device latency; structural sparsity effects on filter counts and memory access patterns are consistent across hardware platforms.

### C.2 — Baseline Training Hyperparameters

| Hyperparameter | ResNet-50 / CIFAR-10 | MobileNetV2-SSD / COCO |
|---|---|---|
| Epochs | 50 | 100 |
| Batch size | 512 | 128 |
| Optimizer | SGD + Momentum 0.9 | SGD + Momentum 0.9 |
| Base LR | 0.04 | 0.04 |
| LR Schedule | Cosine annealing | Cosine annealing |
| Warm-up epochs | 5 | 5 |
| Weight decay | 1e-4 | 1e-4 |
| Input size | 32 × 32 | 300 × 300 |
| Num classes | 10 | 91 (COCO) |

### C.3 — Pruning & Fine-tuning Hyperparameters

After structural pruning, all models were fine-tuned for **5 epochs** using the following settings:

| Hyperparameter | ResNet-50 / CIFAR-10 | MobileNetV2-SSD / COCO |
|---|---|---|
| Fine-tune epochs | 5 | 20 |
| Batch size | 512 | 128 |
| Fine-tune LR | 0.004 (10× reduced) | 0.004 (10× reduced) |
| LR Schedule | Cosine annealing | Cosine annealing |
| Weight decay | 1e-4 | 1e-4 |
| Calibration size | 2 000 samples | 1 000 samples |

### C.3b - End-to-End Pruning Pipeline Timing (ResNet-50 / CIFAR-10)

Measured on a single RTX 4090 at 30% sparsity, averaged across the three methods.

| Pipeline Stage | Approximate Time |
|---|---|
| Importance scoring (calibration) | ~1.5 s (2,000 samples, single forward+backward) |
| Dependency graph construction | ~0.8 s |
| Channel selection & pruning | ~0.5 s |
| Fine-tuning (5 epochs, CIFAR-10) | ~8 min (batch 512, cosine LR 0.004) |
| Total per (method, ratio) config | ~9 min |

## C.4 — Pruning Method Details

**L1-Norm:** Filters ranked by the ■■-norm of their weight tensors. The lowest-ranked filters (per the target ratio) are removed independently per layer, followed by downstream channel correction. No data is required.

**Grad-Act (Ours — Taylor-Made Sparsity):** Importance is computed as the squared product of the gradient and activation at each filter, averaged over a calibration mini-batch. Specifically, for filter $k$ in layer $l$: $I(k,l) = \Sigma\, (\partial L/\partial a \cdot a)^2$ summed over spatial positions and calibration samples. Structural consistency is enforced via DepGraph dependency resolution before removal.

**DepGraph:** Dependency-graph-based structural pruning using magnitude importance (*torch-pruning* library). Groups dependent layers automatically before pruning.

## C.5 — Dataset Details

| Dataset | Task | Train / Val Split | Preprocessing |
|---|---|---|---|
| CIFAR-10 | Image Classification | 50 000 / 10 000 | RandomCrop 32 (pad 4), RandomHFlip, Normalize |
| MS-COCO 2017 | Object Detection | 118 287 / 5 000 | Resize 300×300, Normalize (ImageNet stats) |

**Scope note:** We use CIFAR-10 as a controlled benchmark to isolate pruning behaviour under varying sparsity without the confounding cost of ImageNet-scale training. CIFAR-10's compact spatial resolution amplifies the effect of filter removal, making it well-suited for ablation studies. Extending evaluation to large-scale datasets is identified as future work.

## C.6 — Software Stack

| Component | Version / Details |
|---|---|
| Python | 3.10 |
| PyTorch | 2.x (CUDA 12) |
| torchvision | 0.17+ |
| torch-pruning | 1.x (DepGraph backend) |
| pycocotools | 2.0.x |
| NumPy / Pillow | standard scientific stack |

Total GPU compute used across all experiments (baseline training + pruning + fine-tuning + FPS benchmarks) was approximately **72 GPU-hours** on the RTX 4090.